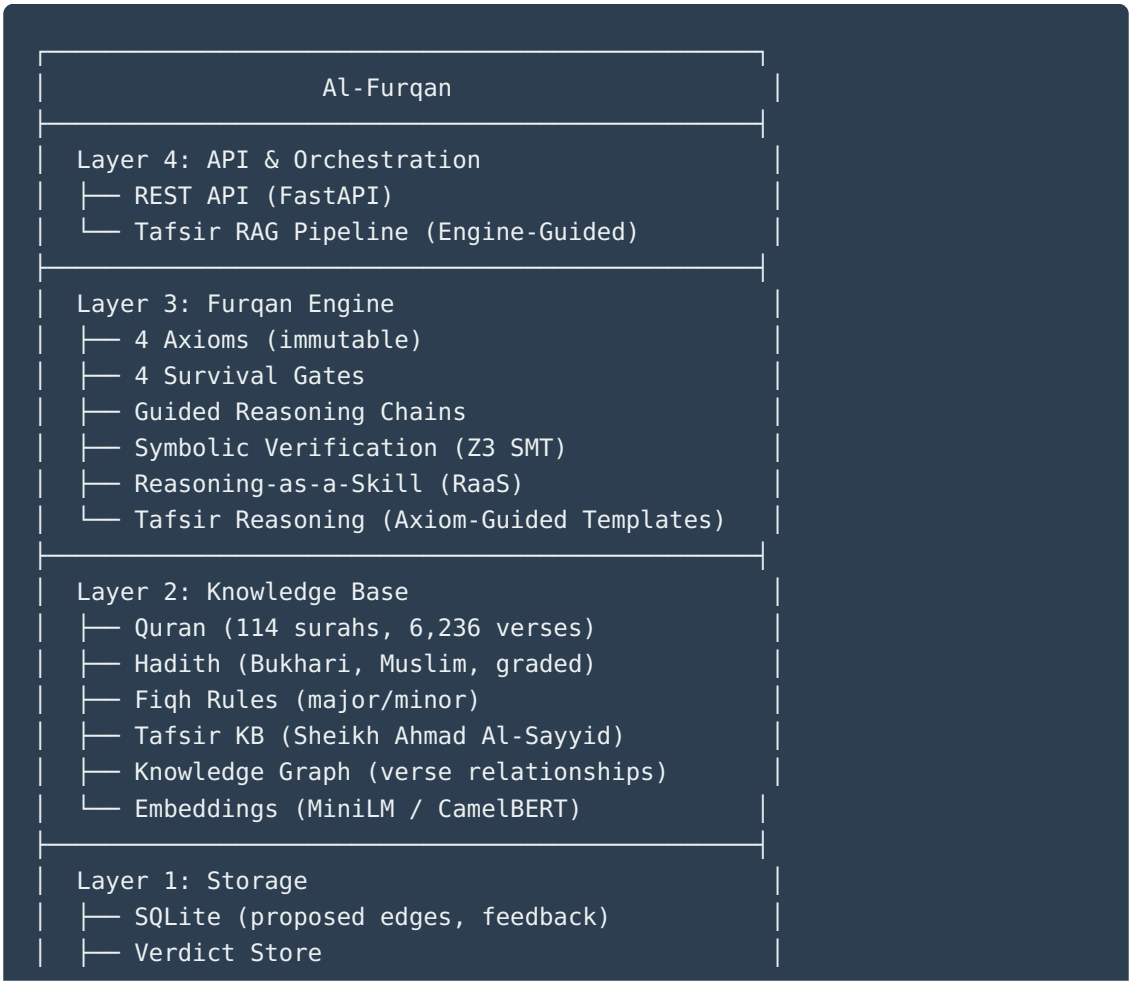
Al-Furqan (الفرقان) — The Criterion

> Axiom-Anchored Neuro-Symbolic Reasoning Engine with Tafsir Knowledge Base

Al-Furqan is a **general-purpose reasoning engine** that evaluates ideas, systems, and claims against immutable logical axioms through formal verification. It uses the Islamic Knowledge Base as its verified reference point — because the axioms themselves establish this as the only source that passes all four survival gates.

Key principle: The LLM speaks. The Code thinks. Z3 proves. The Knowledge Base knows.

Architecture



```
│   └── Feedback Store                                    │
└─────────────────────────────────────────────────────────┘
```

The Four Gates

Every claim passes through 4 sequential survival gates:

Gate	Name	Criterion
1	**Source-Integrity**	Is raw truth preserved without reduction?
2	**Structural-Consistency**	Can the system explain causality without luck?
3	**Mediation-Zeroing**	Is human cognition treated as finite, not authoritative?
4	**Origin-Aware**	Does truth trace to a transcendent source?

Engine-Guided Tafsir RAG Pipeline

A complete pipeline for Quranic reasoning that teaches the LLM **how to think**, not just what to answer:

```
User Question
      ↓
① Query Analyzer          — extracts verses, topics, question type
      ↓
② Reasoning Plan Builder  — LLM selects Axioms & Gates dynamically
      ↓
③ LLM Execution           — thinks + searches KB with tools
      ↓
④ Human Feedback           — 4 verdicts (/// / / )
```

KB Tools (exposed to LLM via function calling):

- `search_kb_by_verse("6:5")` — all edges for a verse
- `search_kb_by_topic("السنة الإلهية")` — semantic topic search

- `search_kb_by_relation("6:5", "LINKED_HADITH")` — specific relation type
- `get_verse_context("6:5", range=3)` — surrounding verses + KB data

Reasoning Templates:

6 templates, each mapped to relevant Axioms + Gates: `TAFSIR` · `VERSE_LINK` · `ISTINBAT` · `COMPARISON` · `SEERAH_LINK` · `GENERAL`

Tafsir Knowledge Base:

- **Source:** الشيخ أحمد السيد — مدارسة سورة الأنعام (23 episodes)
- **Extraction:** Whisper transcription → chunk → LLM extraction → human review
- **Current:** 67 entries (Episode 1), central verses: 6:1, 6:5
- **Structure:** Central verse → linked verses, hadiths, concepts, tafsir

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Symbolic Verification

Z3 SMT solver provides formal mathematical proofs for: - Transcendence Necessity Proof - Final Court Necessity Proof - Axiom consistency verification

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Test Coverage

```
705 tests total
├── Engine (gates, chains, pipeline, axioms, Z3)     ~560
├── Knowledge Base (quran, hadith, fiqh, graph)       ~60
├── Tafsir RAG Pipeline                              100
│   ├── Query Analyzer                                23
│   ├── KB Tools                                      17
│   ├── Tool Executor                                 13
│   ├── Reasoning Plan Builder                        22
│   ├── Pipeline (end-to-end)                         13
│   └── Feedback System                               12
├── RaaS (Reasoning-as-a-Skill)                       31
├── Memory (MCP server)                               56
└── Security (prompt guard, audit, sandbox)           ~30
```

Security

5-layer defense-in-depth: 1. **Prompt Guard** — injection detection + sanitization 2. **Axiom Integrity** — SHA-256 hash verification (immutable) 3. **Output Validator** — response filtering 4. **Adapter Sandbox** — isolated LLM execution 5. **Audit Logger** — hashed question logging

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Project Structure

```
al-furqan/
├── src/al_furqan/
│   ├── engine/                    # Furqan Engine
│   │   ├── axioms.py              # Immutable axioms + gates
│   │   ├── pipeline.py            # Scan → Mirror → Verdict
│   │   ├── chains/                # Guided reasoning chains
│   │   ├── gates/                 # 4 survival gates
│   │   ├── symbolic/              # Z3 formal verification
│   │   ├── security/              # 5-layer security
│   │   └── tafsir/                # Tafsir reasoning pipeline
│   │       ├── axiom_selector.py  # Dynamic axiom/gate selection
│   │       ├── reasoning_plan_builder.py
│   │       ├── reasoning_templates.py
│   │       ├── pipeline.py        # End-to-end RAG pipeline
│   │       └── feedback.py        # Human feedback system
│   ├── kb/                        # Knowledge Base
│   │   ├── collections/           # Quran, Hadith, Fiqh
│   │   ├── graph/                 # Knowledge graph
│   │   ├── ingestion/             # Transcript → KB extraction
│   │   └── tafsir/                # Tafsir KB tools
│   │       ├── query_analyzer.py  # Question analysis
│   │       ├── kb_tools.py        # 4 search tools for LLM
│   │       └── tool_executor.py   # Tool call execution
│   ├── providers/                 # LLM providers (Ollama, DashScope, Anthropic, etc.)
│   ├── store/                     # Verdict + feedback storage
│   └── api/                       # REST API
├── data/
│   ├── quran/                     # Complete Quran text
│   ├── hadith/                    # Hadith collections
│   ├── tafsir/                    # 9 Arabic tafsir books (consolidated)
│   ├── lessons/                   # Transcribed episodes
│   ├── review/                    # Proposed edges (KB)
│   ├── benchmark/                 # Evaluation results
│   └── tafsir_feedback/           # Human feedback entries
├── furqan-raas/                   # Reasoning-as-a-Skill (MCP)
├── furqan-memory/                 # Memory system (MCP)
├── tests/                         # 705 tests
```

```
├── scripts/                    # Processing & evaluation scripts
└── docs/                       # Documentation
```

📚 Documentation

Document	Description
Architecture v3.0	Complete system architecture
RAG Implementation Plan	Engine-Guided RAG pipeline design
Fine-Tuning Plan	SFT + DPO plan for Furqan-27B
Knowledge Graph Docs	KB schema & extraction
Security Policy	5-layer security architecture
RaaS Docs	Reasoning-as-a-Skill
Sprint Report 2026-03-22	Latest sprint (RAG pipeline)

🗺 Roadmap

✅ Done

- [x] Furqan Engine (4 gates, reasoning chains, Z3 verification)
- [x] Knowledge Base (Quran, Hadith, Fiqh, 9 Tafsir books)
- [x] Knowledge Graph (transcript extraction, central verse tracking)
- [x] Security (5-layer defense, axiom integrity)
- [x] Engine-Guided RAG Pipeline (query analysis → reasoning plan → LLM + tools → feedback)
- [x] Human Feedback System (4 verdicts, full context storage)
- [x] Dynamic Axiom/Gate Selection (LLM chooses from raw Engine definitions)
- [x] 705 tests passing

Next

- [ ] Download & process episodes 2-23 (KB expansion)
- [ ] Collect 500+ human-reviewed pipeline responses
- [ ] Semantic search (FAISS + embeddings)
- [ ] API endpoint for the pipeline

Future

- [ ] Fine-tune Furqan-27B (SFT + DPO on Qwen3.5-27B-Claude-Opus-Distilled)
- [ ] Local deployment (vLLM/SGLang)
- [ ] Multi-scholar KB support
- [ ] QLP v3.0 integration (Local-First)

Tech Stack

Layer	Technology
Language	Python 3.12
LLM Providers	DashScope (Qwen), Anthropic, Ollama, OpenAI-compatible
Formal Verification	Z3 SMT Solver
Embeddings	MiniLM / CamelBERT
Storage	SQLite (WAL mode)
API	FastAPI
Testing	pytest (705 tests)
Transcription	OpenAI Whisper
Future Training	LLaMA-Factory + Unsloth (LoRA/QLoRA)

Team

1. **ماجد عارف (Magid Arif)** — CEO
2. **محمود السمان (Mahmoud Al-Samman)** — Principal Software Architect
3. **محمد الأشماوي (Muhammad Al-Ashmawy)** — Research Lead & Domain Expert
4. **آية أبو الوفا (Aya Abu Al-Wafa)** — Researcher
5. **مصطفى مرزوق (Mustafa Marzouk)** — Researcher
6. **علي (Ali)** — Researcher
7. **مريم ياسر (Maryam Yasser)** — Researcher

License

Al-Furqan: The Code thinks. The LLM speaks. Z3 proves. The Knowledge Base knows.